

S7CAS-PLEIADES-CARTE-MERE

Pleiades

01

Revision 1.00

Date : 13/01/2026

Revision history	

Package size conversion	
Metric	Imperial
1005	0402
1608	0603
2012	0805
3216	1206
3225	1210
6432	2512

Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group		Revision		
11x17	01		1.00		
Date	13/01/2026		Sheet	1 of 13	
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_TITLE.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

Arriver 24 V

VFS VBATT

3V3

VCC1V5 VCCO_3

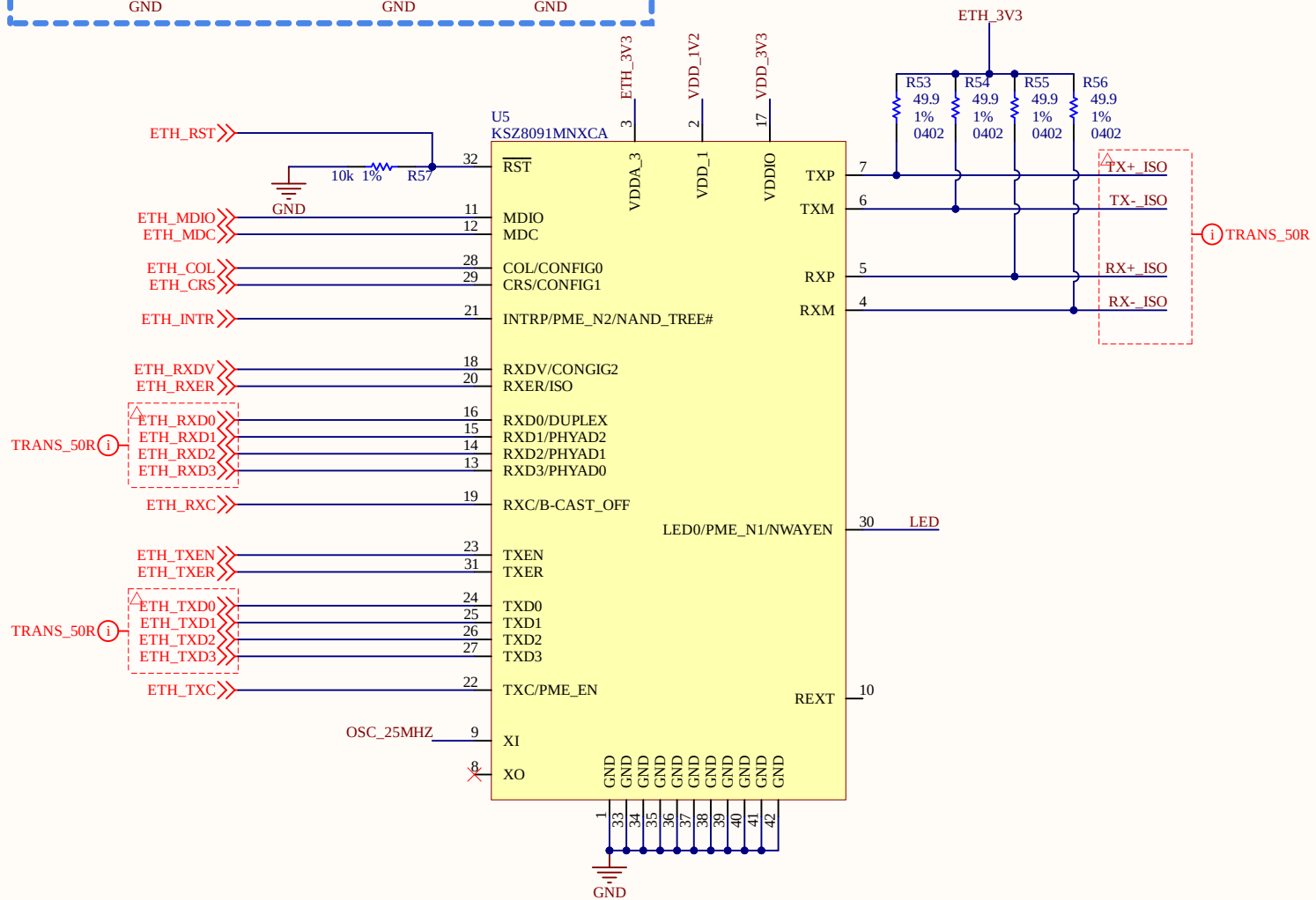
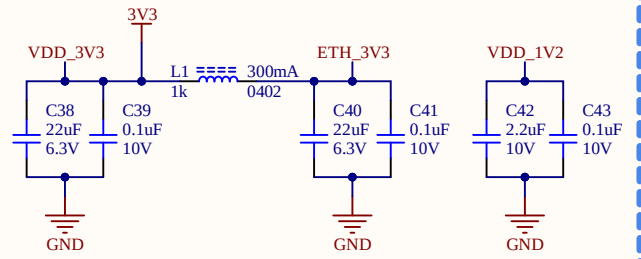
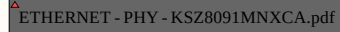
VCCO_0 VCCO_1 VCCO_2

Avoir un entrée de 3V3 nommer 3V3
VCCO 1.5V pour le DDR3

Pas oublier les PGOOD et EN pour l'alimentation du DDR3

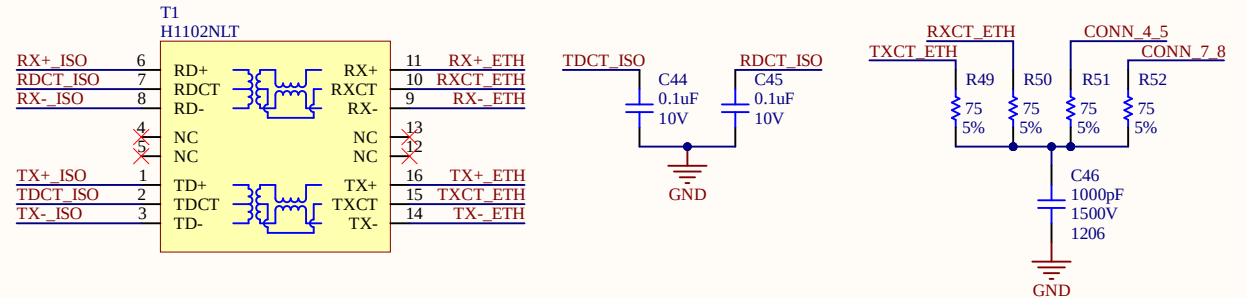
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Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	2 of 13
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_ALIM.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

ETHERNET ALIMENTATION

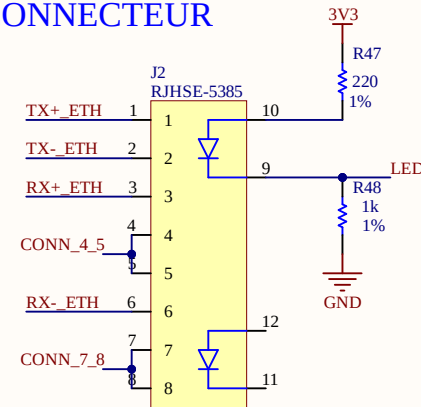


Ferite de 1k?
Valider les condensateur (Surtout pour ETH_3V3)
Faut tu mettre transmission entre Iso et Connecteur

ISO



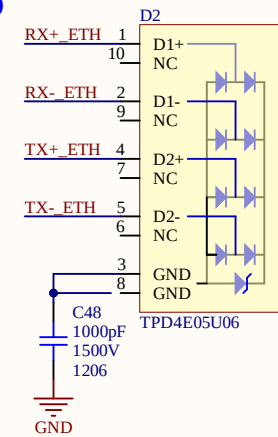
CONNECTEUR



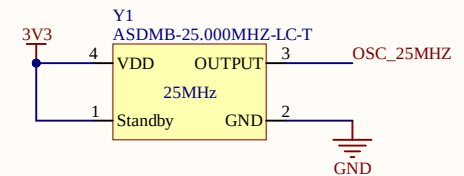
Pin #	Wire Color Legend	Signal
1	 White/Green	TX+
2	 Green	TX-
3	 White/Orange	RX+
4	 Blue	TRD2+
5	 White/Blue	TRD2-
6	 Orange	RX-
7	 White/Brown	TRS3+
8	 Brown	TRD3-

- Isolation du ETHERNET 1.5kV
- Ajout d'un condensateur de 1.5kV
- Ligne de transmission de 50R

ESD

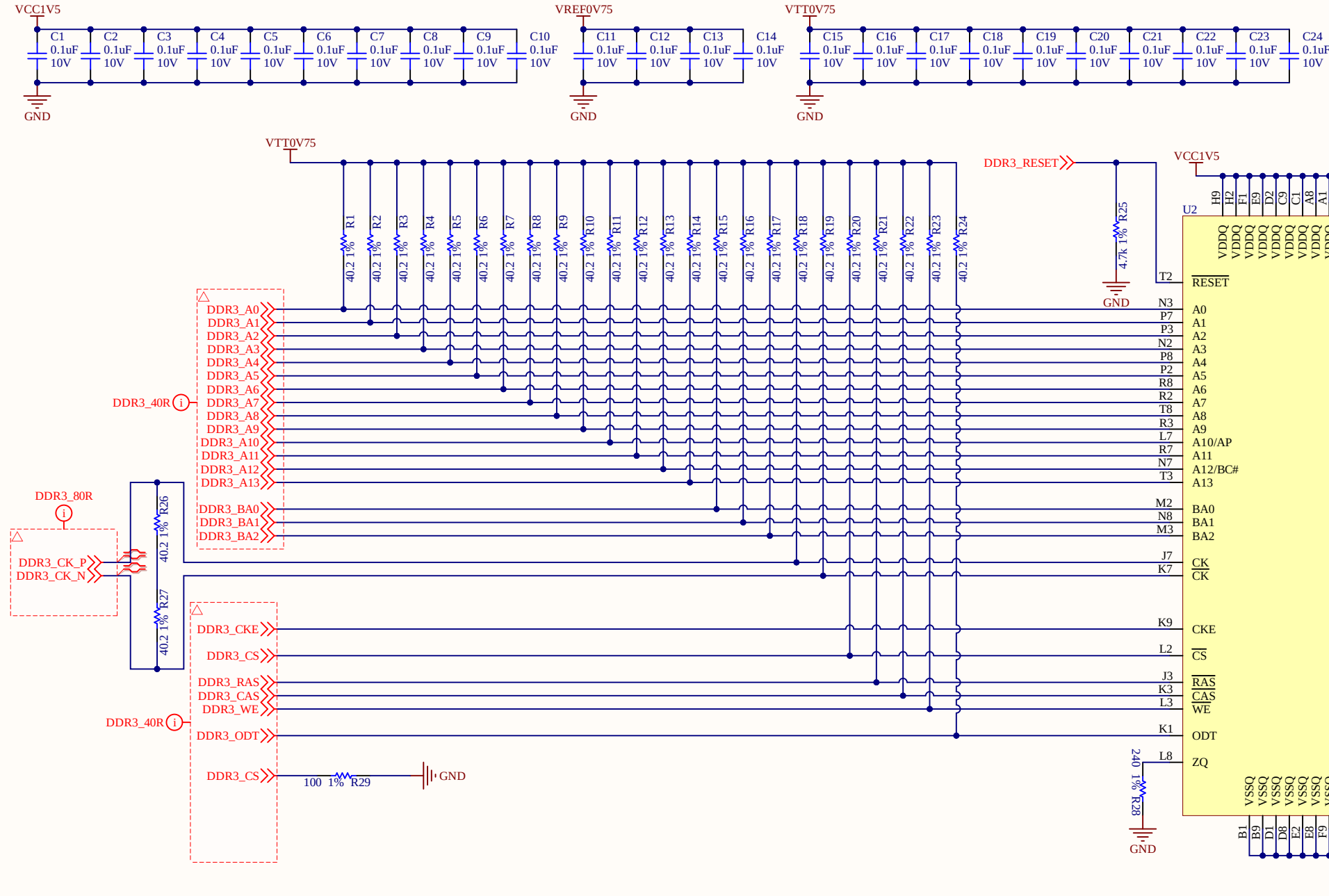


OSCILLI



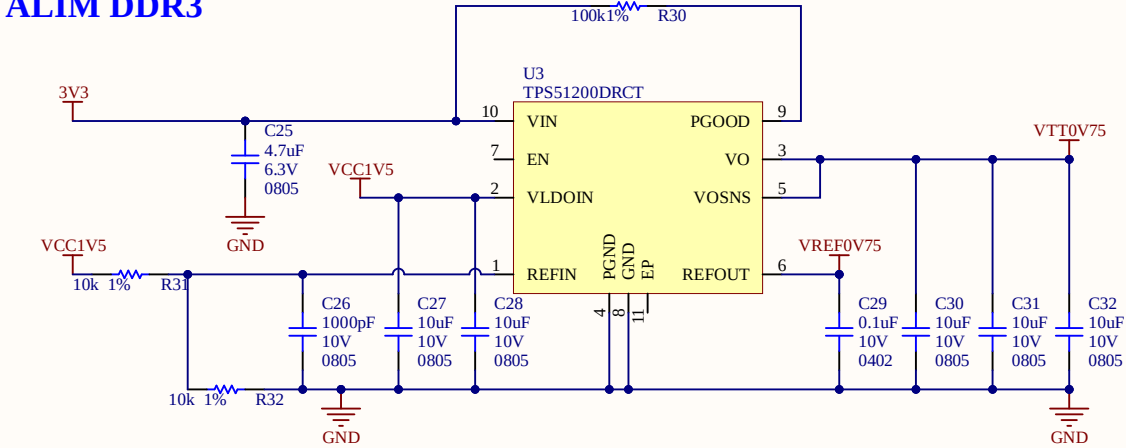
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Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group		Revision		
11x17		01	1.00		
Date	13/01/2026	Sheet	3	of	13
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_ETHERNET.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA



MANQUE ZIO?
ANALYSER LES COURANTS POUR LE VTT

ALIM DDR3

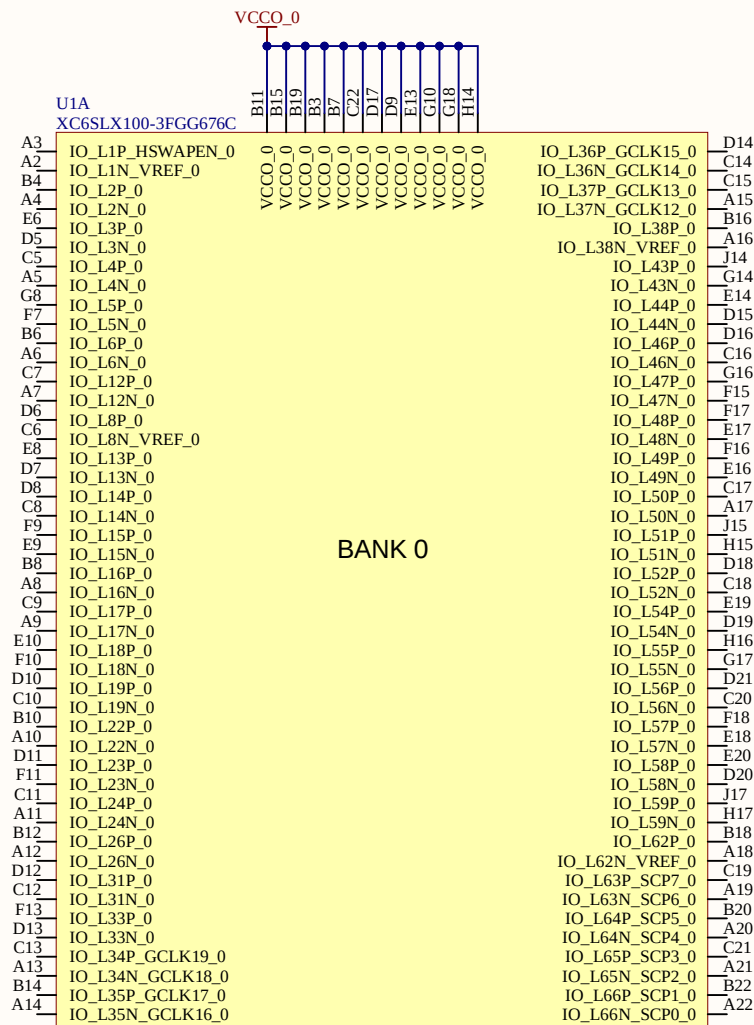


-Ligne de transmission de 40R
-Resistance parallèle sur la ligne de transmission de 40.2Ohms
-Clock en pair différentiel 80R

Sortie PGOOD?
EN?

Sheet Name			DDR3		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	11x17	Group	01		Revision
13/01/2026			Sheet	4	of 13
Filename			S7CAS-PLEIADES-CARTE-MERE_P01_DDR3.SchDoc		Designers
			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA

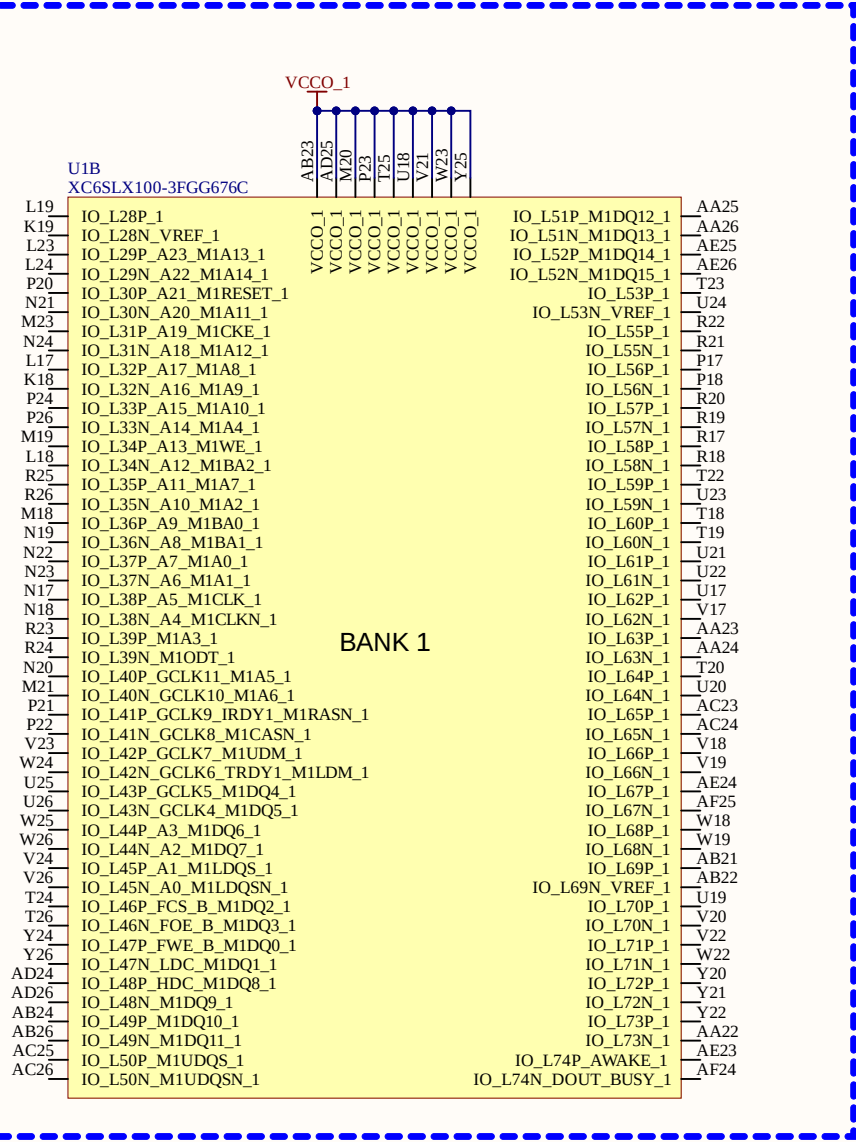


Sheet Name			FPGA - BANK 0		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
Date	13/01/2026		Sheet	5	of 13
Filename				Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK0.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

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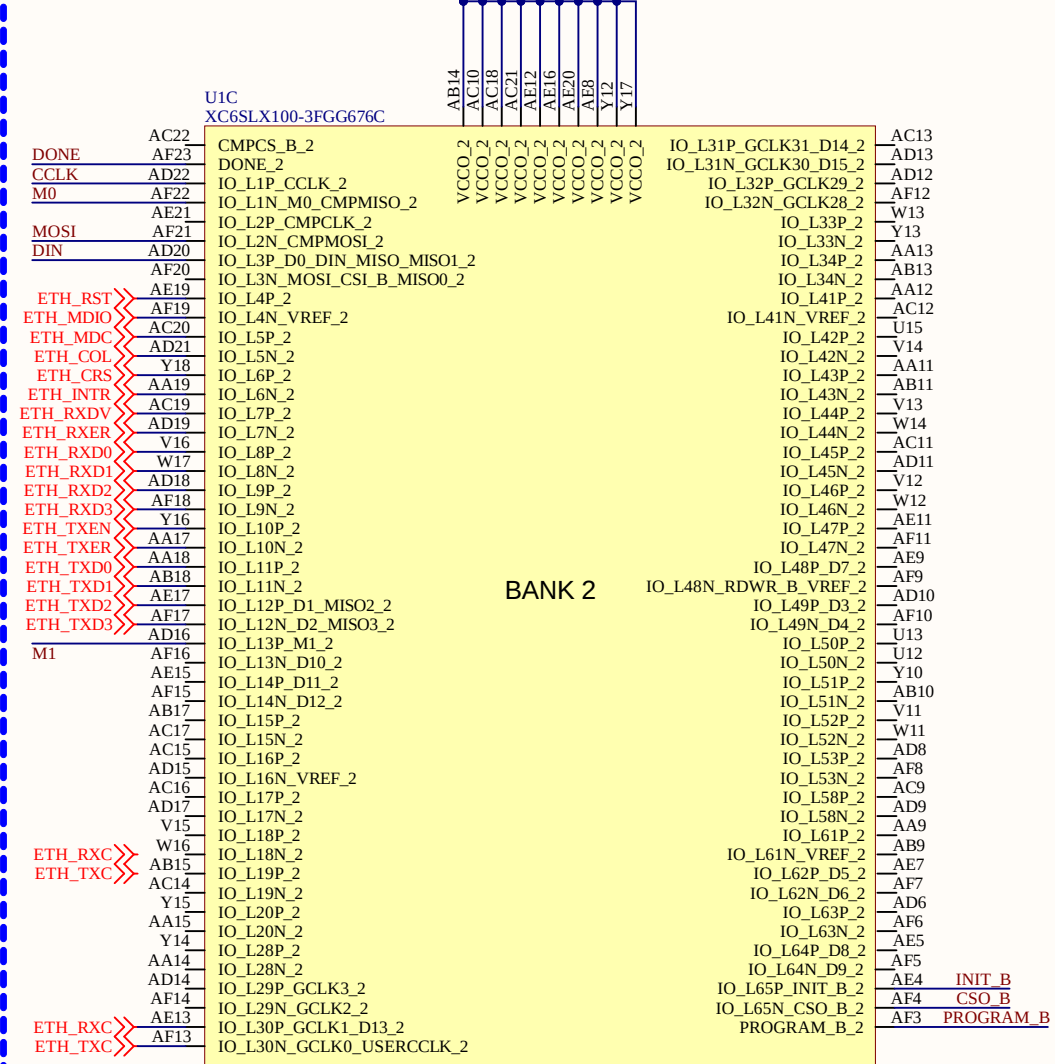
BANK 1

Pin	Function	Pin	Function
L19	IO_L28P_1	AA25	IO_L51P_M1DQ12_1
K19	IO_L28N_VREF_1	AA26	IO_L51N_M1DQ13_1
L23	IO_L29P_A23_M1A13_1	AE25	IO_L52P_M1DQ14_1
L24	IO_L29N_A22_M1A14_1	AE26	IO_L52N_M1DQ15_1
P20	IO_L30P_A21_M1RESET_1	T23	IO_L53P_1
N21	IO_L30N_A20_M1A11_1	U24	IO_L53N_VREF_1
M23	IO_L31P_A19_M1CKE_1	R22	IO_L55P_1
N24	IO_L31N_A18_M1A12_1	R21	IO_L55N_1
L17	IO_L32P_A17_M1A8_1	P17	IO_L56P_1
K18	IO_L32N_A16_M1A9_1	P18	IO_L56N_1
P24	IO_L33P_A15_M1A10_1	R20	IO_L57P_1
P26	IO_L33N_A14_M1A4_1	R19	IO_L57N_1
M19	IO_L34P_A13_M1WE_1	R17	IO_L58P_1
L18	IO_L34N_A12_M1BA2_1	R18	IO_L58N_1
R25	IO_L35P_A11_M1A7_1	T22	IO_L59P_1
R26	IO_L35N_A10_M1A2_1	U23	IO_L59N_1
M18	IO_L36P_A9_M1BA0_1	T18	IO_L60P_1
N19	IO_L36N_A8_M1BA1_1	T19	IO_L60N_1
N22	IO_L37P_A7_M1A0_1	U21	IO_L61P_1
N23	IO_L37N_A6_M1A1_1	U22	IO_L61N_1
N17	IO_L38P_A5_M1CLK_1	U17	IO_L62P_1
N18	IO_L38N_A4_M1CLKN_1	V17	IO_L62N_1
R23	IO_L39P_M1A3_1	AA23	IO_L63P_1
R24	IO_L39N_M1ODT_1	AA24	IO_L63N_1
N20	IO_L40P_GCLK11_M1A5_1	T20	IO_L64P_1
M21	IO_L40N_GCLK10_M1A6_1	U20	IO_L64N_1
P21	IO_L41P_GCLK9_IRDY1_M1RASN_1	AC23	IO_L65P_1
P22	IO_L41N_GCLK8_M1CASN_1	AC24	IO_L65N_1
V23	IO_L42P_GCLK7_M1UDM_1	V18	IO_L66P_1
W24	IO_L42N_GCLK6_TRDY1_M1LDM_1	V19	IO_L66N_1
U25	IO_L43P_GCLK5_M1DQ4_1	AE24	IO_L67P_1
U26	IO_L43N_GCLK4_M1DQ5_1	AF25	IO_L67N_1
W25	IO_L44P_A3_M1DQ6_1	W18	IO_L68P_1
W26	IO_L44N_A2_M1DQ7_1	W19	IO_L68N_1
V24	IO_L45P_A1_M1LDQS_1	AB21	IO_L69P_1
V26	IO_L45N_A0_M1LDQSN_1	AB22	IO_L69N_VREF_1
T24	IO_L46P_FCS_B_M1DQ2_1	U19	IO_L70P_1
T26	IO_L46N_FOE_B_M1DQ3_1	V20	IO_L70N_1
Y24	IO_L47P_FWE_B_M1DQ0_1	V22	IO_L71P_1
Y26	IO_L47N_LDC_M1DQ1_1	W22	IO_L71N_1
AD24	IO_L48P_HDC_M1DQ8_1	Y20	IO_L72P_1
AD26	IO_L48N_M1DQ9_1	Y21	IO_L72N_1
AB24	IO_L49P_M1DQ10_1	Y22	IO_L73P_1
AB26	IO_L49N_M1DQ11_1	AA22	IO_L73N_1
AC25	IO_L50P_M1UDQS_1	AE23	IO_L74P_AWAKE_1
AC26	IO_L50N_M1UDQSN_1	AF24	IO_L74N_DOUT_BUSY_1

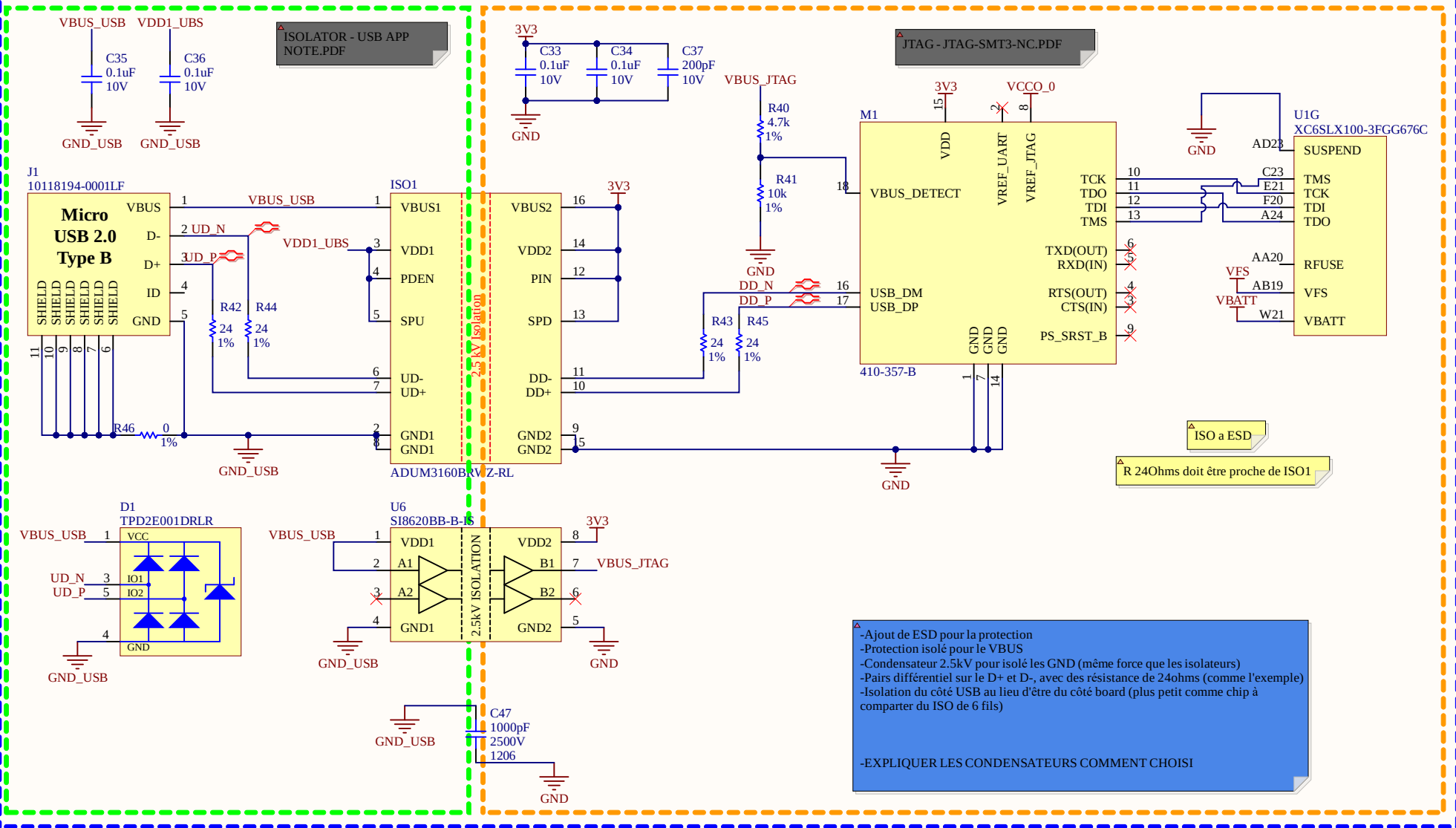


Sheet Name			FPGA - BANK 1		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group	Revision			
11x17	01	1.00			
Date	13/01/2026	Sheet	6	of	13
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK1.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

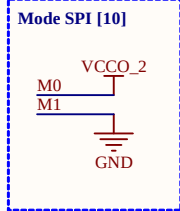
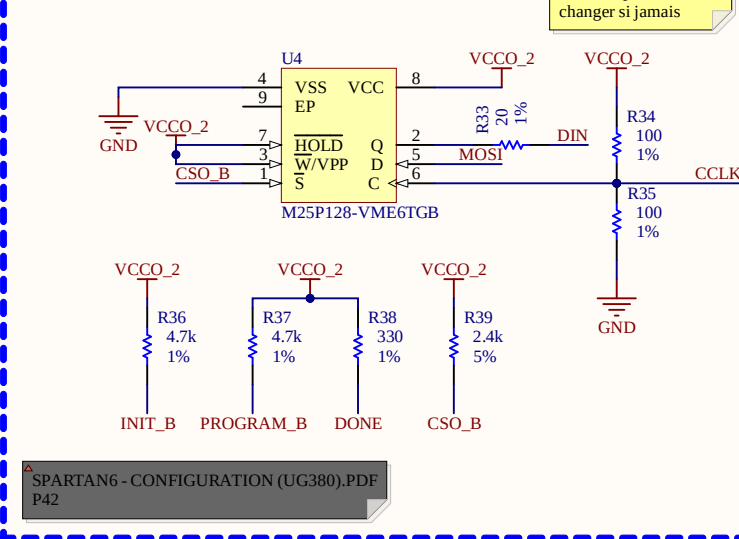
FPGA



USB



FLASH

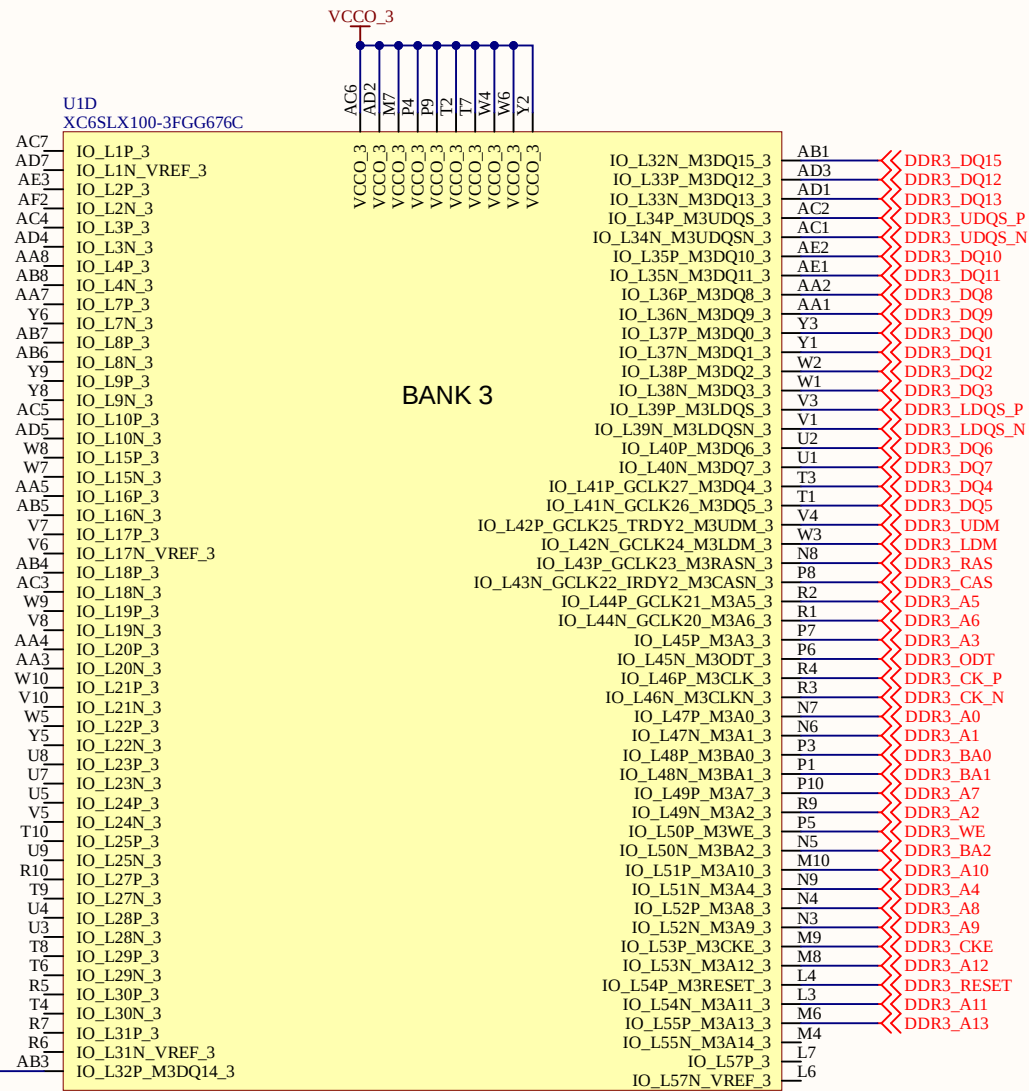


-M0 et M1 choisi pour avoir le mode SPI
- Ajout d'une résistance de 20 ohms sur Q

Mettre condensateur sur TVS sur l'alim

Sheet Name			FPGA - BANK 2		
Project Title			S7CAS-PLIADES-CARTE-MERE		
Global Project			Pleiades		
Size	11x17	Group	01		Revision
Date		13/01/2026		Sheet	7 of 13
Filename		S7CAS-PLIADES-CARTE-MERE_P01_FPGA-BANK2.SchDoc		Designers	
				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

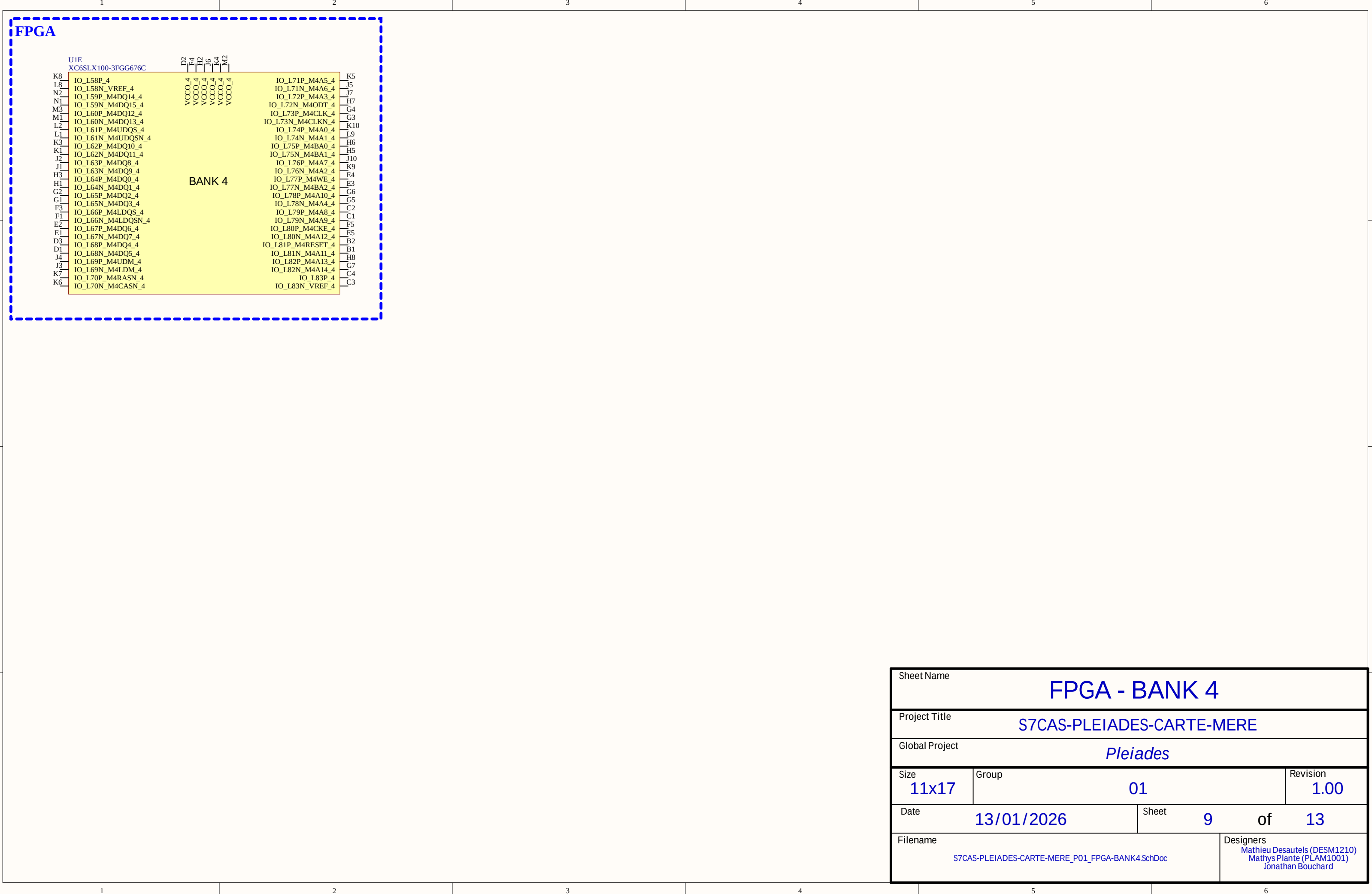
FPGA

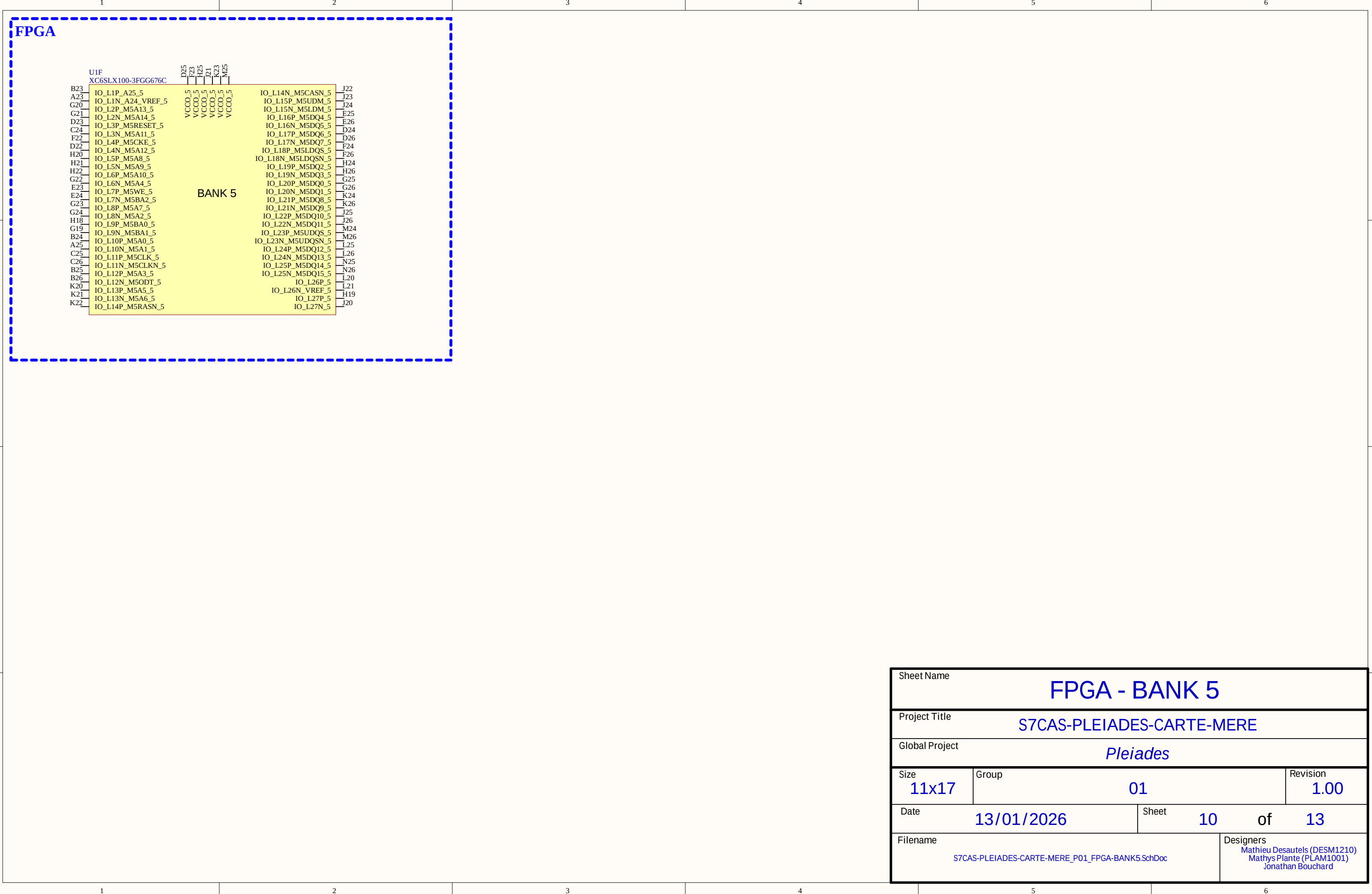


Rajouter les VREF avec des condensateurs pour le FPGA

VREF0V75

Sheet Name			FPGA - BANK 3		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
Date	13/01/2026		Sheet	8	of 13
Filename				Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK3.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	





1	2	3	4	5	6
A					A
B					B
C					C
D					D

Sheet Name		CONNECTORS			
Project Title		S7CAS-PLEIADES-CARTE-MERE			
Global Project		Pleiades			
Size	Group	Revision			
11x17	01	1.00			
Date	13/01/2026	Sheet	12	of	13
Filename		Designers			
S7CAS-PLEIADES-CARTE-MERE_P01_CONNECTORS.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard			

1	2	3	4	5	6
A					
B					
C					
D					

Sheet Name		ASIC	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	13 of 13
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_ASIC.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	